IDEATION PHASE

Electric Vehicle Market Performance & Insights

Project Documentation | Tableau Data Analytics

Project Title	Electric Vehicle Market Performance & Insights
Project Type	Data Analytics and Interactive Dashboard
Primary Tool	Tableau Public

1. Project Overview

The Electric Vehicle Market Performance & Insights project aims to explore electric vehicle (EV) market data and turn it into clear, actionable insights through an interactive Tableau dashboard and story. The analysis focuses on market performance, manufacturer comparison, regional distribution, vehicle characteristics, pricing, battery capacity, driving range, and customer ratings where available in the dataset.

2. PS — Problem Statement

The electric vehicle market is growing and becoming more diverse, with differences across manufacturers, vehicle categories, regions, prices, battery capacities, and driving ranges. However, these details can be difficult to compare when they are spread across a large dataset or viewed as raw tables. Stakeholders may struggle to identify leading manufacturers, understand sales trends, compare vehicle specifications, and spot patterns in price and performance.

Problem statement: How can EV market data be analysed and presented through an interactive, easy-to-understand dashboard so that users can compare market performance, identify important trends, and make more informed decisions about EV manufacturers, regions, and vehicle characteristics?

3. Target Users and Their Needs

Target User	Primary Need
Business and market analysts	Compare market performance, manufacturers, and regional patterns.
EV manufacturers and strategy teams	Understand competitive positioning and vehicle attributes.
Potential buyers and EV enthusiasts	Explore price, battery capacity, driving range, and ratings where available.
Students and researchers	Explore EV-market patterns and communicate findings using visual analytics.

4. Ideation — Possible Ideas / Solutions

- **Idea 1: Interactive EV Market Overview Dashboard —** A high-level dashboard with key performance indicators and filters to summarise the dataset.

- **Idea 2: Manufacturer and Sales Comparison —** Visuals comparing manufacturers, sales trends, and category-wise performance.
- **Idea 3: Regional EV Market Exploration —** A map and regional comparison charts to reveal geographic patterns, where the data supports location-level analysis.
- **Idea 4: EV Specification and Price Analysis —** Scatter plots and distribution charts to explore relationships among price, battery capacity, driving range, and ratings.
- **Idea 5: EV Market Story / Guided Insights —** A sequence of dashboard views that explains the main findings and supports a clear narrative.

5. Idea Prioritization

The ideas were compared using four criteria: user value, feasibility with the available dataset and Tableau, clarity of insights, and portfolio/demo value. Each criterion is scored from 1 (low) to 5 (high). Weights: User Value 35%, Feasibility 25%, Insight Clarity 25%, and Portfolio Value 15%. Scores are preliminary planning estimates and should be validated against the actual data.

Idea	User Value (35%)	Feasibility (25%)	Insight Clarity (25%)	Portfolio Value (15%)	Weighted Score / 5
1. Market Overview Dashboard	5	5	5	5	5.00
2. Manufacturer & Sales Comparison	5	5	5	4	4.85
3. Regional Market Exploration	4	4	4	4	4.00
4. Specification & Price Analysis	4	4	4	4	4.00
5. Guided Tableau Story	4	5	5	5	4.65

Prioritization outcome: The highest-priority solution is an interactive EV Market Overview Dashboard, supported by manufacturer and sales comparisons, specification/price analysis, and a guided Tableau Story.

6. Selected Idea and Rationale

Selected idea: Develop an interactive Tableau dashboard titled “Electric Vehicle Market Performance & Insights” and accompany it with a guided Story. The dashboard will combine KPI cards, trend analysis, manufacturer rankings, category and regional comparisons, and vehicle-specification analysis. It was selected because it brings multiple views together in one place, is feasible with Tableau, and creates a clear portfolio artifact.

7. Scope for Later Phases

Detailed data cleaning and field validation will be documented in Requirement Analysis / Project Planning as appropriate. Dashboard layout and chart-level design will be covered in Project Design. Implementation steps will be covered in Project Development, while results and checks will be recorded in Performance Testing and Final Outcome. This avoids repeating the same details in every project file.